



Getting Started

CORE

`install.packages("ivdtools")`

Install ivdtools from CRAN

`library(ivdtools)`

Load the package and register all analysis and plotting methods

`packageVersion("ivdtools")`

Confirm the installed ivdtools version

`help("regression", package = "ivdtools")`

Open function arguments, return values, and examples

`# Recommended: .qmd, .Rmd, or Jupyter`

Use an interactive, reproducible analysis document

`sessionInfo()`

Record the analysis environment for reproducibility

Data Checks, Summaries & Grouping

PREP

`outliers_test(d, "result", "grubbs")`

Flag candidate outliers with Grubbs, ESD, Dixon, or IQR

`normal_test(d, "result", "auto")`

Assess normality with SW, AD, Lillie, or CvM

`plot(nt, type = c("qq", "his"))`

Inspect the distribution with Q-Q plots or histograms

`replicate_to_mean(raw, "sample", c("new", "ref"))`

Summarize replicates as mean, SD, and n

`split_by_cutpoints(d, "ref", c(100, 500))`

Split data at predefined cutpoints

`split_by_factors(d, "lot")`

Split data by lot, site, or another factor

`continuous_to_binary(d, "marker", 10)`

Convert continuous values to binary outcomes

Bottle / Lot Differences

EP15

`ba <- bottle_anova(d, value ~ bottle)`

Assess mean differences for a single bottle or lot factor

`bottle_anova(d, value ~ lot/vial)`

Assess nested effects such as bottles within lots

`tukey(ba, term = "bottle")`

Run Tukey pairwise comparisons for a significant factor

`p <- precision(d, value ~ bottle); variance(p)`

Partition between-bottle and within-bottle variance

Method Comparison

EP09

`obj <- mcr(d, "id", "candidate", "reference")`

Create a paired candidate-versus-reference object

`obj <- describe(obj)`

Calculate method-specific location, spread, and range

`obj <- correlation(obj, "pearson")`

Calculate Pearson, Spearman, or Kendall correlation

`obj <- regression(obj, method = "deming", lambda = 1)`

Fit OLS, WLS, Deming, or Passing-Bablok regression

`obj <- bland_altman(obj, type = "difference")`

Estimate the mean difference and limits of agreement

`obj <- outlier(obj, type = "difference")`

Identify candidate outliers in paired differences

`obj <- bias(obj, mdl = c(15, 25), interval = "both")`

Estimate bias and intervals at medical decision levels

`predict(obj, reference = c(15, 25))`

Predict candidate results from reference values

`plot(obj, type = "regression", interval = "confidence")`

Plot scatter, regression, Bland-Altman, or bias views

`summary(obj)`

Summarize completed analysis steps and results

Linearity

EP06

`linearity_replicates(imprecision = 4, adl = 5, levels = 8)`

Calculate replicates per level from imprecision and ADL

`linearity_panel(seq(0, 1, .2), high = 1000, low = 10)`

Generate a high/low mixture panel for linearity

`linearity_endpoints(lloq=10,uloq=1000,cv_high=5,cv_at_lloq=20)`

Derive linearity endpoints from LLoQ, ULoQ, and CV targets

`lin <- linearity(d,"level","y","expected",weights="level_variance",adl_rel=5)`

Fit linear or polynomial models and assess nonlinearity

`plot(lin, type = "fit")`

Plot fit, replicates, precision, residuals, or differences

Precision

EP05 / EP15

`p <- precision(d, result ~ day/run, by = "sample")`

Create a repeatability and within-laboratory precision object

`p <- outlier(p, method = "grubbs")`

Flag candidate outliers within sample or concentration groups

`p <- normal(p, method = "auto")`

Assess normality within sample groups

`p <- variance(p, method = "auto")`

Estimate variance components with ANOVA or REML

`vc(p)`

Extract variance, SD, CV, and component shares

`p <- ci(p)`

Calculate confidence intervals for SD and CV

`p <- profile(p, model.no = 1:10)`

Fit a Sadler precision profile across concentration

`report(p, day = "day", run = "run", table = "CI")`

Build a standardized precision report by day, run, and sample

`plot(p, type = "var")`

Plot dot, histogram, variance, Q-Q, or profile views

`list_sadler()`

List available Sadler profile models

Reference Intervals

EP28

`ri <- reference_interval(d,"value",method="all",id="id",seed=1728)`

Estimate reference limits and confidence intervals

`plot(ri)`

Plot the reference distribution and interval limits

Sample Size

PLANNING

`sample_size_bland_altman(mu = .2, sd = 1, delta = 2.5)`

Estimate sample size for LoA precision or power

`sample_size_proportion_ci(p = .3, half_width = .05)`

Size a one-proportion confidence interval by target half-width

`sample_size_proportion(p0 = .2, p1 = .35)`

Size a one-sample proportion test against a target

Calibration / Response Curves

CURVE FIT

`list_equation(category = "Response")`

List built-in response curves and equations

`fit <- fit_equation("4PLC", d, x = "Con", y = "OD")`

Fit a calibration response curve with a built-in equation

`fit_equation(y ~ a * (1 - exp(-b*x)), d)`

Fit a user-defined nonlinear equation

`compare_equation(d, x = "Con", y = "OD", eqs = c("4PLC", "5PLC"))`

Compare candidate equations with AIC and diagnostics

`coef(fit); residuals(fit)`

Extract equation parameters and residuals

`predict(fit, newdata, inverse = TRUE)`

Back-calculate unknown concentrations from responses

`plot(fit, interval = "confidence")`

Plot the fitted curve, observations, and confidence interval

Stability

EP25

`stability_plan(allowable_drift = 5, variability = 1.5, power = .9)`

Plan time points from allowable drift, variability, and power

`sb <- stability_bias(d,"cond","time","y",reference_condition="ref",limit=5)`

Estimate bias versus the reference condition at each time point

`sg <- stability_regression(d,"time","y","cond","compare","test","ref",limit=5)`

Fit stability trends and calculate one-sided confidence limits

`stability_time(sg, limit = 5, max_time = 365)`

Solve when a trend or confidence limit first reaches the acceptance limit

`arrhenius(d, "temp", "time", "result", target_temp = 5, limit = 10)`

Extrapolate stability at the target temperature with Arrhenius models

`mkt(d, c("probe1", "probe2"), "time")`

Calculate mean kinetic temperature from time-temperature records

Reference Material Bias

EP09

`reference_bias(d, "result", "reference", "u_ref", level = "level")`

Evaluate reference-material bias with reference uncertainty



ROC, AUC & Cutoffs

EP24

r <-
roc(d,c("m1","m2"),"reference",id="id",positive="case")
Create a subject-level ROC analysis object

r <- describe(r)
Summarize case/control counts and marker distributions

r <- auc(r, cols = c("m1", "m2"))
Calculate AUC, standard error, and confidence interval

r <- cutoff(r)
Return candidate cutoffs and their performance metrics

r <- mlr(r, cols = c("m1", "m2"), name = "panel")
Combine multiple markers with logistic regression

auc_compare(r, curves = c("m1", "m2"), method = "delong")
Compare paired ROC curves with DeLong or related methods

predict(r, mlr = "panel", newdata = new_d)
Predict probabilities for new samples

plot(r, curves = c("m1", "m2"), cutpoint = "Youden")
Plot ROC curves or cutoff performance

summary(r)
Summarize completed ROC steps and key results

Qualitative Agreement

EP12

tab <-
raw_to_table(d,"candidate","reference",id="id",positive="positive")
Create a contingency table from paired raw classifications

tab <- counts_to_table(tp = 45, fp = 3, tn = 97, fn = 5)
Create a contingency table from TP, FP, TN, and FN counts

describe(tab)
Summarize counts, sample size, and class margins

diagnostics(tab, ci.method = "wilson")
Calculate sensitivity, specificity, predictive values, and LRs

kappa(tab)
Estimate Cohen's kappa and its confidence interval

mcnemar(tab)
Assess paired disagreement with McNemar's test

summary(tab)
Summarize agreement metrics, intervals, and paired tests

LoD / LoQ

EP17

lob <- sensitivity_lob(blank, "result", "sample", lot = "lot")
Estimate LoB by nonparametric, parametric, or zero-LoB methods

lod <- sensitivity_lod(low,"y","sample",lob=lob,method="classical")
Estimate LoD by classical, precision-profile, or probit methods

loq <- sensitivity_loq(d,"y","sample","assigned",goal=.20)
Estimate LoQ from Westgard, RMS, or precision targets

sensitivity_verify(d,"y","sample",limit="lod",claim=2,lob=lob)
Verify a LoB, LoD, or LoQ claim with success rates and one-sided intervals

sensitivity_design("classical",d,"y","sample",limit="lod")
Compare sensitivity-study design requirements with observed counts

plot(lod)
Plot LoB, LoD, or LoQ estimates and diagnostics

Interference

EP07 / YY/T 1789.5

interference_replicates(sd = 1.5, detectable_difference = 3)
Estimate replicates per group from repeatability and detectable difference

ip <- interference_paired(d,"y",condition="condition",by="analyte")
Compare paired interferent and control preparations

dr <- interference_dose_response(d,"y","concentration",by="analyte")
Estimate interference bias across interferent concentration

predict(dr, target = 10, metric = "percent")
Interpolate the concentration producing a target effect

pt <- interference_patient(d,"id","group","method","y","test","control","new","ref")
Confirm interference with patient samples and a reference method

plot(dr, type = "percent")
Plot results, absolute bias, or relative bias

Dilution, Spiking & Hook Effect

EP34

dr <- dilution_recovery(d,"sample","y","factor",assigned="assigned")
Calculate dilution recovery and intervals at each factor

plot(dr, type = "summary")
Plot dilution recovery, summary, or observed values

spike_concentration(100, 0.9, 1000, 0.1)
Convert spike concentration by purity and molecular-weight balance

sr <- spike_recovery(d,"sample","y","cond","spike","solvent","added")
Correct solvent contribution and estimate spike recovery

hk <- hook_effect(d,"concentration","response",series="series",uloq=1000)
Identify high-dose hook effects and downturn limits

plot(sr); plot(hk)
Plot spike recovery or the raw hook-response curve

Commutability

EP14 / EP30 / IFCC

cm <- commutability(d,"id","type","method","y",approach="ep14")
Evaluate commutability with the EP14 prediction-interval approach

commutability(d,"id","type","method","y",approach="ep30",reference_procedure="ref")
Evaluate commutability with the EP30 reference-procedure framework

commutability(d,"id","type","method","y",approach="ifcc",criterion=2)
Evaluate commutability with the IFCC Part 2 bias-difference approach

summary(cm); plot(cm)
Summarize design checks, classifications, and diagnostic plots

C5 / C50 / C95

EP12

cc <- c5_c95(d,"signal","level",cutoff=20,direction="geq")
Estimate concentrations at 5%, 50%, and 95% detection probability

plot(cc, type = "probability")
Plot the detection-probability curve or precision profile

Quality Control

WESTGARD

list_westgard(brief = FALSE)
List supported Westgard quality-control rules

qc <- qc_chart(d,"value",mean=100,sd=2,group="level",run="run")
Create a Levey-Jennings QC object and evaluate rules

plot(qc)
Plot the QC series and mark rule violations

yd <- youden_plot(d, "sample1", "sample2", mean1 = 100, mean2 = 50)
Use a two-level Youden plot to inspect systematic and random error

plot(yd)
Display random- and systematic-error patterns in a Youden plot

Measurement Uncertainty

GUM

uA <- uncertainty_type_a(x, "repeatability", quantity = "mean")
Estimate a Type A standard uncertainty from replicate data

uB <- uncertainty_type_b("certificate", 100, uncertainty = 2, k = 2)
Evaluate a Type B component from certificates, resolution, or other evidence

uncertainty_iqc(d, "result", run = "run")
Estimate a precision component from long-term internal QC

uncertainty_homogeneity(d, "bottle", "result")
Estimate between-bottle, between-lot, or between-unit homogeneity

uncertainty_stability(d, "time", "result", shelf_life = 365)
Estimate a time-related uncertainty component from stability data

uncertainty_characterization(d, "laboratory", "result")
Evaluate assigned-value uncertainty from multi-laboratory results

ub <- uncertainty_combine(uA, uB, value = 100)
Combine independent components into standard or expanded uncertainty

uncertainty_propagate(model,components,"monte_carlo",seed=42)
Propagate a measurement model by Delta or Monte Carlo methods

uncertainty_traceability(stages, value = 100)
Combine uncertainty budgets across the traceability chain

plot(ub)
Plot each uncertainty component's contribution to the total budget

ivdtools 0.2.5 · MIT · github.com/hiox-tech/ivdtools

2 / 2